VII SEMESTER (PROFESSIONAL ELECTIVE-IV)	L	T	P	C
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20AM7E03 - NOSQL DATABASES				

COURSE OBJECTIVE:
The student will be able to

1. Define, compare and use the four types of NoSQL Databases (Document-oriented, Key-Value Pairs, Column-oriented and Graph).
2. Demonstrate an understanding of the detailed architecture, define objects, load data, query data and performance tune Column-oriented NoSQL databases.
3. Explain the detailed architecture, define objects, load data, query data and performance tune Document-oriented NoSQL databases.

COURSE OUTCOMES:
At the end of the course, student will be able to
CO1: Discuss about Aggregate Data Models
CO2: Explain about Master-Slave Replication, Peer-to-Peer Replication
CO3: Describe the Structure of Data, Scaling, Suitable Use Cases
CO4: Make use of Complex Transactions Spanning Different Operations
CO5: Identify Routing, Dispatch and Location-Based Services

UNIT-I:
Why NoSQL, The Value of Relational Databases, Getting at Persistent Data, Concurrency, Integration, A (Mostly) Standard Model, Impedance Mismatch, Application and Integration Databases, Attack of the Clusters, The Emergence of NoSQL, Aggregate Data Models; Aggregates, Example of Relations and Aggregates, Consequences of Aggregate Orientation, Key-Value and Document Data Models, Column-Family Stores, Summarizing Aggregate-Oriented Databases. More Details on Data Models; Relationships, Graph Databases, Schema less Databases, Materialized Views, Modelling for Data Access.

UNIT-II:
Distribution Models: Single Server, Shading, Master-Slave Replication, Peer-to-Peer Replication, Combining Shading and Replication. Consistency, Update Consistency, Read Consistency, Relaxing Consistency, The CAP Theorem, Relaxing Durability, Quorums. Version Stamps, Business and System Transactions, Version Stamps on Multiple Nodes.

UNIT-III:
What Is a Key-Value Store, Key-Value Store Features, Consistency, Transactions, Query Features, Structure of Data, Scaling, Suitable Use Cases, Storing Session Information, User Profiles, Preference, Shopping Cart Data, When Not to Use, Relationships among Data, Multi operation Transactions, Query by Data, Operations by Sets.

UNIT-IV:
Document Databases, What Is a Document Database, Features, Consistency, Transactions, Availability, Query Features, Scaling, Suitable Use Cases, Event Logging, Content Management Systems, Blogging Platforms, Web Analytics or Real-Time Analytics, Ecommerce Applications, When Not to Use, Complex Transactions Spanning different Operations, Queries against Varying Aggregate Structure.

UNIT-V:
Graph Databases, What Is a Graph Database, Features, Consistency, Transactions, Availability, Query Features, Scaling, Suitable Use Cases, Connected Data, Routing, Dispatch and Location-Based Services, Recommendation Engines, When Not to Use.

Text Books:

1. Sadalage, P. & Fowler, No SQL Distilled: A Brief Guide to the Emerging world of polyglot persistence, Pearson Addision Wesley,2012.

Reference Books:

1. Dan Sullivan, "NoSQL for Mere Mortals", 1st Edition Pearson Education India, 2015.(ISBN 13: 978-9332557338)
2. Dan McCreary and Ann Kelly, " Making Sense of NoSQL; A guide for managers and the rest of us", 1st Edition, Manning Publication- Dreamtech press, 2013. (ISBN-13:978-9351192022).
3. Kristina Chodorow, " Mongodb: The Definitive Guide-Power full and scalable data storage", 2nd Edition, O'Reilly publications,2013. (ISBN-13:978-9351102694).